



Question Bank

E-gov Question Bank 2080



Tribhuvan University Institute of Science and Technology 2080

Bachelor Level / Science

Full Marks: 60 + 20 + 20

Bsc. CSIT

Pass Marks: 24 + 8 + 8

E-Governance

Time: 3 Hours

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Section A

Attempt any two questions.

1. Differentiate between e-governance and e-government. Explain the issues in e-governance implementations and how to solve these issues? ✓

2. What do you mean by data warehousing and data mining? Explain the applications of data warehousing ✓



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Best Answer of the Question

Differentiate between e-governance and e-government. Explain the issues in e-governance implementations and how to solve these issues?

Expert approved answer

E-Governance and e-Government are related concepts but have distinct meanings:

1. **E-Government:** E-government refers to the utilization of Information and Communication Technologies (ICTs) by government agencies to transform and enhance the efficiency, effectiveness, transparency, and accountability of governmental operations, service delivery, and interactions with citizens, businesses, and other government entities. It primarily focuses on the use of technology within the government itself to streamline processes, improve decision-making, and provide better services to citizens.
2. **E-Governance:** E-governance encompasses a broader scope and refers to the overall framework of processes, structures,



[VIEW ALL ANSWERS](#)



government and its stakeholders including



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Best Answer of the Question

2. **E-Governance:** E-governance encompasses a broader scope and refers to the overall framework of processes, structures, mechanisms, and relationships that facilitate the interaction between government and its stakeholders, including citizens, businesses, civil society organizations, and other government entities. It encompasses not only the use of ICTs within the government (e-government) but also their utilization in governance processes, policymaking, public participation, and service delivery.

Issues in E-Governance Implementations and Solutions:

1. **Digital Divide:** One of the major challenges in e-governance is the digital divide, which refers to the gap between those who have access to and can effectively use ICTs and those who do not. To address this issue, governments can implement initiatives to increase digital literacy and provide affordable access to technology, such as community centers with internet access and training programs.
2. **Privacy and Security Concerns:** E-



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Best Answer of the Question

2. **Privacy and Security Concerns:** E-governance involves the collection, storage, and processing of vast amounts of data, raising concerns about privacy and security. Governments need to implement robust cybersecurity measures, data protection laws, and mechanisms for ensuring transparency and accountability in data handling and processing.
3. **Infrastructure and Connectivity:** In many regions, especially rural and remote areas, inadequate ICT infrastructure and connectivity pose significant barriers to e-governance initiatives. Governments can address this issue by investing in expanding broadband infrastructure, leveraging mobile technologies, and implementing alternative solutions such as offline modes for accessing e-government services.
4. **Capacity Building and Change Management:** Implementing e-governance requires building the capacity of government officials to effectively utilize ICTs and adapt to new processes and technologies. Comprehensive training programs and change management strategies can help facilitate the adoption



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Best Answer of the Question

ICTs and adapt to new processes and technologies. Comprehensive training programs and change management strategies can help facilitate the adoption of e-governance initiatives and ensure their successful implementation.

5. **Interoperability and Standardization:** Lack of interoperability and standardization among different government systems and databases can hinder the seamless integration of e-governance initiatives. Governments should adopt open standards and interoperability frameworks to facilitate data exchange and integration across different departments and agencies.
6. **Citizen Engagement and Participation:** Effective e-governance requires meaningful citizen engagement and participation in decision-making processes. Governments can leverage ICTs to create platforms for citizen feedback, consultation, and collaboration, fostering greater transparency, accountability, and trust in government institutions.



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Best Answer of the Question

What do you mean by data warehousing and data mining? Explain the applications of data warehousing and data mining in census and prices of essential commodities.

Expert approved answer

Data Warehousing:

Data warehousing is the process of collecting, storing, and managing large volumes of data from various sources within an organization. The data is structured, cleaned, and transformed to facilitate easy access and analysis. The primary goal of data warehousing is to provide a centralized repository of integrated data for decision-making and business intelligence purposes.

Data Mining:

Data mining is the process of discovering

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Best Answer of the Question

Data mining is the process of discovering patterns, trends, and insights from large datasets using various statistical and machine learning techniques. It involves analyzing the data to uncover hidden patterns and relationships that can be used to make predictions, optimize processes, and gain actionable insights.

Applications in Census:

1. **Data Warehousing:** In the context of a census, data warehousing would involve collecting and storing demographic information, such as population counts, age distribution, household characteristics, and socioeconomic indicators, from various sources such as surveys, administrative records, and census forms. This data is then organized and stored in a centralized data warehouse, allowing census authorities to access and analyze the data efficiently.
2. **Data Mining:** Data mining techniques can be applied to census data to extract valuable insights and patterns. For



[VIEW ALL ANSWERS](#)



based on various attributes such as



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Best Answer of the Question

2. **Data Mining:** Data mining techniques can be applied to census data to extract valuable insights and patterns. For example, clustering algorithms can group similar demographic regions together based on various attributes, such as population density or income levels. Association rule mining can identify relationships between different demographic factors, such as the correlation between education levels and employment rates.

Applications in Prices of Essential Commodities:

1. **Data Warehousing:** For essential commodities pricing, data warehousing would involve collecting and storing data related to commodity prices, supply chain information, market demand, weather patterns, and other relevant factors. This data is then integrated and stored in a centralized warehouse, providing a comprehensive view of the factors influencing commodity prices.
2. **Data Mining:** Data mining techniques can be applied to analyze pricing data and identify patterns and trends. For example,



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historical data and market trends. Market



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Applications in Prices of Essential Commodities:

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- 2. Data Mining:** Data mining techniques can be applied to analyze pricing data and identify patterns and trends. For example, time series analysis can be used to forecast future commodity prices based on historical data and market trends. Market basket analysis can uncover patterns of co-occurring commodity purchases, helping retailers optimize pricing strategies and promotions.



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Best Answer of the Question

Explain broadcasting/wider dissemination model and comparative analysis model of e-governance with practical example.

Expert approved answer

Broadcasting/Wider Dissemination Model

In the broadcasting or wider dissemination model of e-governance, the government disseminates information, services, and policies to citizens in a unidirectional manner, typically through digital channels such as websites, social media, email newsletters, and mobile applications. The focus is on providing easy access to information and services to a broad audience without active engagement or interaction.

Practical Example: A government launches a website where citizens can access information about public services, such as



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website where citizens can access information about public services, such as applying for permits, paying taxes, or accessing healthcare services. The website provides detailed information about eligibility criteria, required documents, and application procedures, allowing citizens to access services conveniently from their homes or mobile devices without the need to visit government offices.

Comparative Analysis Model

In the comparative analysis model of e-governance, the government utilizes technology to collect, analyze, and compare data from different sources to make informed decisions and improve governance processes. This model emphasizes data-driven decision-making and performance evaluation through comparative analysis of various indicators, benchmarks, and best practices.

Practical Example: A government implements

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Collects data on key performance indicators



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Best Answer of the Question

This model emphasizes data-driven decision-making and performance evaluation through comparative analysis of various indicators, benchmarks, and best practices.

Practical Example: A government implements a performance management system that collects data on key performance indicators (KPIs) related to different aspects of governance, such as service delivery, infrastructure development, and fiscal management. The system allows government agencies to compare their performance against predefined benchmarks and identify areas for improvement. For instance, agencies can compare their efficiency in processing permit applications or their responsiveness to citizen complaints with similar agencies in other regions or countries to identify best practices and implement reforms accordingly.



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Best Answer of the Question

Explain in brief about scope and content of e-governance.

Expert approved answer

The scope and content of e-governance encompass a wide range of activities, processes, and initiatives aimed at leveraging information and communication technologies (ICTs) to transform and improve governance processes, service delivery, and citizen engagement. Here's a brief overview of the scope and content of e-governance:

1. **Service Delivery:** E-governance aims to enhance the delivery of government services to citizens, businesses, and other stakeholders through digital channels. This includes services such as applying for permits and licenses, paying taxes, accessing healthcare and education services, and receiving government benefits and assistance.

2



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accountability in government operations by



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benefits and assistance.

2. **Transparency and Accountability:** E-governance promotes transparency and accountability in government operations by providing access to information about government policies, decisions, and expenditures. This includes initiatives such as open data portals, online budget transparency platforms, and citizen participation mechanisms.
3. **Citizen Engagement:** E-governance facilitates citizen engagement and participation in governance processes by providing platforms for feedback, consultation, and collaboration. This includes online forums, social media channels, and mobile applications that enable citizens to voice their opinions, report issues, and contribute to decision-making processes.
4. **Administrative Efficiency:** E-governance aims to improve the efficiency and effectiveness of government operations by automating processes, reducing paperwork, and streamlining workflows. This includes initiatives such as electronic document management systems, online



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This includes initiatives such as electronic document management systems, online procurement platforms, and digital workflow automation tools.

5. **Policy Development and Implementation:**

E-governance supports the development and implementation of government policies and programs through data-driven decision-making, performance monitoring, and impact assessment. This includes the use of data analytics, predictive modeling, and evaluation frameworks to inform policy formulation and evaluation.

6. **Interoperability and Integration:**

E-governance involves ensuring interoperability and integration among different government systems, databases, and services to facilitate seamless data exchange and collaboration. This includes adopting open standards, interoperability frameworks, and data exchange protocols to enable integration across different departments and agencies.

7. **Capacity Building and Skills Development:**

E-governance requires building the capacity of government officials, IT professionals, and other stakeholders to effectively utilize



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Skill development initiatives, and



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Best Answer of the Question

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6. **Interoperability and Integration:** E-governance involves ensuring interoperability and integration among different government systems, databases, and services to facilitate seamless data exchange and collaboration. This includes adopting open standards, interoperability frameworks, and data exchange protocols to enable integration across different departments and agencies.
7. **Capacity Building and Skills Development:** E-governance requires building the capacity of government officials, IT professionals, and other stakeholders to effectively utilize ICTs and adapt to new technologies and processes. This includes training programs, skill development initiatives, and knowledge sharing platforms to enhance digital literacy and competency.



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Best Answer of the Question

Discuss the important role of cyberlaw in Nepal.

Expert approved answer

Cyberlaw plays a crucial role in Nepal, as it provides the legal framework for regulating activities and addressing issues related to cyberspace, including electronic transactions, online privacy, cybersecurity, digital rights, and cybercrimes. Here are some important roles of cyberlaw in Nepal:

1. Regulation of Electronic Transactions:

Cyberlaw in Nepal provides legal recognition and validity to electronic transactions, contracts, and signatures. It establishes the rules and procedures for conducting business transactions electronically, promoting e-commerce and digital economy growth.

2. Protection of Online Privacy: Cyberlaw in Nepal includes provisions for protecting the privacy and personal data of individuals online. It regulates the collection, use, and



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OTHER ENTITIES OPERATING IN CYBERSPACE



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Best Answer of the Question

Nepal includes provisions for protecting the privacy and personal data of individuals online. It regulates the collection, use, and disclosure of personal information by businesses, government agencies, and other entities operating in cyberspace.

3. Cybersecurity Regulations: Cyberlaw in Nepal addresses cybersecurity concerns by establishing regulations and standards for securing information systems, networks, and critical infrastructure. It mandates measures such as data encryption, access controls, incident reporting, and cybersecurity audits to protect against cyber threats and attacks.

4. Prevention and Prosecution of Cybercrimes: Cyberlaw in Nepal defines various cybercrimes, such as hacking, cyberbullying, online fraud, identity theft, and unauthorized access to computer systems. It outlines the penalties and legal procedures for investigating, prosecuting, and punishing offenders engaged in illegal activities in cyberspace.

5. Promotion of Digital Rights: Cyberlaw in Nepal protects digital rights and freedoms, including freedom of expression, access to



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and use the internet without censorship or



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Nepal protects digital rights and freedoms, including freedom of expression, access to information, and online participation. It safeguards individuals' rights to access and use the internet without censorship or undue restrictions from government authorities.

6. **Consumer Protection in E-commerce:**

Cyberlaw in Nepal includes provisions for consumer protection in e-commerce transactions. It requires businesses to provide accurate product information, secure payment processing, and fair dispute resolution mechanisms to ensure consumer confidence and trust in online shopping.

7. **International Cooperation and**

Collaboration: Cyberlaw in Nepal facilitates international cooperation and collaboration on cyber issues, including information sharing, mutual legal assistance, and extradition of cybercriminals. It enables Nepal to align its cyber laws and regulations with international standards and best practices.

8. **Capacity Building and Awareness:**

Cyberlaw in Nepal promotes capacity



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8. **Capacity Building and Awareness:**

Cyberlaw in Nepal promotes capacity building initiatives and public awareness campaigns to educate individuals, businesses, government agencies, and law enforcement authorities about cyber risks, best practices for cybersecurity, and legal rights and obligations in cyberspace.



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What do you mean by maturity levels of e-governance?

Expert approved answer

Level 1 – Initial (E-Government)

At this level, the focus is on moving services online for user convenience and cost savings, but data and its uses are siloed and extremely limited. “If the organizational view is that a high percentage of online services or mobile access represents a modern digital government, then more education and advocacy is needed to show what real digital government looks like, and its benefits,” said Di Maio. “To make the case for advancement, create case studies explaining how digital transformation will ease or remove high-priority pain points for the organization.”

Level 2 – Developing (Open)



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Level 2 — Developing (Open)

Level 2 is not necessarily after level 1. E-government and open government programs often coexist, with different leadership and priorities. Open government often takes the form of public-facing programs intended to promote transparency, citizen engagement and the data economy. Examples we see today are nascent open data initiatives, often in the context of smart city programs such as the Copenhagen Data Exchange.

Level 3 — Defined (Data-Centric)

On this level the focus shifts from simply listening to citizen or user needs to proactively exploring the new possibilities inherent in strategically collecting and leveraging data. The key performance indicators here are “how much of our data is open?” and “how many of our applications are built on open data?” It’s tempting at this point to engage in vanity

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Best Answer of the Question

much of our data is open, and how many of our applications are built on open data?" It's tempting at this point to engage in vanity projects or skip ahead before the proper groundwork is laid; it's paramount to remain focused on designing and implementing data-centric strategies and processes.

Level 4 — Managed (Fully Digital)

By this level, the organization, agency or department has fully committed to a data-centric approach to improving government, and the preferred approach to innovation is based on open data principles. Data flows regularly across organizational boundaries, leading to easier interactions and better services for constituents. It's possible at this stage to encounter privacy-related backlashes, as citizens can be uncomfortable with how their data is being collected and used.

Therefore, it is important to ensure that data is used within existing norms and regulations, and that this is communicated.



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Therefore, it is important to ensure that data is used within existing norms and regulations, and that this is communicated.

Level 5 – Optimizing (Smart)

At this point, the process of digital innovation using open data is embedded deeply across the entire government, with buy in and leadership from the top tier of policymakers. The innovation process is predictable and repeatable, even in the face of disruptions or sudden events that require rapid responses.



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Best Answer of the Question

Discuss the e-Readiness assessment of infrastructure preparedness.

Expert approved answer

E-readiness assessment of infrastructure preparedness refers to the evaluation of a country or organization's readiness and capability to adopt and utilize information and communication technologies (ICTs) effectively. This assessment focuses specifically on the state of ICT infrastructure, including telecommunications networks, internet connectivity, hardware, and software resources, necessary to support e-governance, e-commerce, and digital transformation initiatives. Here are key aspects of e-readiness assessment of infrastructure preparedness:

1. Telecommunications Infrastructure: This aspect assesses the availability, reliability,



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Best Answer of the Question

1. **Telecommunications Infrastructure:** This aspect assesses the availability, reliability, and quality of telecommunications infrastructure, including landline, mobile, and broadband networks. It examines factors such as network coverage, bandwidth capacity, and availability of high-speed internet access in urban, rural, and remote areas.
2. **Internet Connectivity:** E-readiness assessment evaluates the accessibility and affordability of internet connectivity for individuals, businesses, and government agencies. It considers factors such as internet penetration rates, cost of internet services, and availability of public Wi-Fi hotspots and internet cafes.
3. **Hardware Resources:** This aspect focuses on the availability and affordability of hardware resources such as computers, smartphones, tablets, and other devices necessary for accessing and utilizing digital services and applications. It assesses the penetration rates of ICT devices among different segments of the population and the availability of affordable computing devices in the market.

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population and the availability of affordable computing devices in the market.

4. **Software Resources:** E-readiness assessment examines the availability and accessibility of software resources, including operating systems, productivity software, and specialized applications required for e-governance, e-commerce, and other digital activities. It assesses the availability of licensed software, open-source solutions, and compatibility with different hardware platforms.
5. **Digital Literacy and Skills:** In addition to infrastructure readiness, e-readiness assessment also considers the level of digital literacy and skills among the population, government officials, and workforce. It evaluates the availability of training programs, educational resources, and capacity-building initiatives to enhance digital literacy and skills development.
6. **Regulatory Environment:** E-readiness assessment examines the regulatory environment governing ICT infrastructure development and deployment, including policies related to telecommunications licensing, spectrum allocation, competition, and investment incentives. It assesses the



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competition in the ICT sector



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assessment examines the regulatory environment governing ICT infrastructure development and deployment, including policies related to telecommunications licensing, spectrum allocation, competition, and investment incentives. It assesses the effectiveness of regulatory frameworks in promoting investment, innovation, and competition in the ICT sector.

7. **Cybersecurity Preparedness:** This aspect evaluates the level of cybersecurity preparedness to protect ICT infrastructure from cyber threats and attacks. It assesses the availability of cybersecurity policies, regulations, and mechanisms for monitoring, detecting, and responding to cyber incidents and breaches.
8. **Interoperability and Standards:** E-readiness assessment considers the interoperability of ICT infrastructure and adherence to international standards and best practices. It evaluates the compatibility and integration of different hardware and software components to ensure seamless data exchange and interoperability across systems and platforms.



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Best Answer of the Question

Mention the existing scenario of e-governance in Nepal.

Expert approved answer

Nepal has been making efforts to enhance its e-governance capabilities, but the progress has been somewhat slow and faced challenges. Here's an overview of the existing scenario of e-governance in Nepal:

- 1. Digital Infrastructure:** Nepal has been gradually improving its digital infrastructure, including telecommunications networks and internet connectivity. However, access to high-speed internet remains limited, especially in rural and remote areas. Efforts are underway to expand broadband coverage and improve digital connectivity across the country.
- 2. Government Portals and Services:** The Government of Nepal has developed several online portals and platforms to



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portals for citizen registration, tax filing,



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Government of Nepal has developed several online portals and platforms to provide citizens with access to government services and information. These include portals for citizen registration, tax filing, passport and visa applications, land records, and various other administrative services. However, the availability and usability of these portals may vary, and some services may still require physical visits to government offices.

3. **Legal Framework:** Nepal has enacted legislation and policies to support e-governance initiatives and regulate electronic transactions, cybersecurity, and data protection. However, implementation and enforcement of these laws may face challenges, and there may be a need for further reforms and capacity building in this regard.
4. **Citizen Engagement:** There have been efforts to promote citizen engagement and participation in governance processes through digital channels. Social media platforms and online forums are increasingly being used to gather feedback, solicit opinions, and involve citizens in decision-making processes. However, there



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solicit opinions, and involve citizens in decision-making processes. However, there is still room for improvement in terms of the effectiveness and responsiveness of these mechanisms.

5. **Capacity Building:** Capacity building initiatives and training programs have been conducted to enhance digital literacy and skills among government officials, IT professionals, and the general public. These efforts aim to build the necessary human capital to support e-governance initiatives and utilize ICTs effectively.
6. **Challenges:** Despite these efforts, Nepal faces several challenges in advancing its e-governance agenda. These include limited internet penetration and digital literacy, inadequate ICT infrastructure in rural areas, budget constraints, bureaucratic hurdles, and concerns related to cybersecurity and data privacy.
7. **Future Directions:** Nepal has recognized the importance of e-governance in improving government efficiency, transparency, and service delivery. Moving forward, there is a need for sustained investment in digital infrastructure,



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challenges and realize the full potential of



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7. **Future Directions:** Nepal has recognized the importance of e-governance in improving government efficiency, transparency, and service delivery. Moving forward, there is a need for sustained investment in digital infrastructure, capacity building, regulatory reforms, and citizen engagement to overcome existing challenges and realize the full potential of e-governance in Nepal.



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Best Answer of the Question

What are different IT tools that are used to deliver services to citizens.

Expert approved answer

There are various IT tools used to deliver services to citizens, especially in the realm of e-government and digital governance. Here are some examples:

1. **Online Portals:** Government websites or portals where citizens can access information, forms, and services online.
2. **Mobile Apps:** Governments often develop mobile applications that citizens can use to access services, receive notifications, and interact with government agencies.
3. **E-Government Platforms:** Comprehensive platforms that integrate various government services into a single interface for citizens.
4. **Digital Payment Systems:** Online payment



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4. **Digital Payment Systems:** Online payment gateways and systems that allow citizens to pay for government services and transactions electronically.
5. **Identity Management Systems:** Secure systems for managing citizen identities, such as digital IDs or biometric authentication, which are crucial for accessing online services securely.
6. **Customer Relationship Management (CRM) Software:** Tools that help government agencies manage interactions and communications with citizens more effectively.
7. **Document Management Systems:** Software for storing, organizing, and managing documents and records related to citizen interactions and government processes.
8. **Geographic Information Systems (GIS):** Tools for mapping and spatial analysis, used by governments for various purposes including urban planning, emergency response, and environmental management.

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9. **Data Analytics and Business Intelligence Tools:** Software for analyzing data collected from citizens to gain insights, improve services, and make data-driven decisions.
10. **Social Media Platforms:** Governments often use social media channels to communicate with citizens, share information, and gather feedback.
11. **Online Surveys and Feedback Systems:** Platforms for collecting feedback and opinions from citizens about government services and policies.
12. **Chatbots and Virtual Assistants:** Automated systems that can interact with citizens online, answer common questions, and assist with basic transactions.
13. **Open Data Portals:** Platforms where governments publish datasets and information publicly for citizens to access and use for various purposes, such as research and analysis.
14. **Remote Collaboration Tools:** Software for enabling remote work and collaboration



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services



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11. **Online Surveys and Feedback Systems:**

Platforms for collecting feedback and opinions from citizens about government services and policies.

12. **Chatbots and Virtual Assistants:**

Automated systems that can interact with citizens online, answer common questions, and assist with basic transactions.

13. **Open Data Portals:** Platforms where governments publish datasets and information publicly for citizens to access and use for various purposes, such as research and analysis.

14. **Remote Collaboration Tools:** Software for enabling remote work and collaboration among government employees, which can improve efficiency and accessibility of services.

15. **Cybersecurity Solutions:** Tools and measures to ensure the security and integrity of government IT systems and protect citizen data from cyber threats.



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Best Answer of the Question

Discuss about the e-government security architecture.

Expert approved answer

E-government security architecture refers to the framework and measures put in place to safeguard electronic government systems, data, and services from various cyber threats and vulnerabilities. It encompasses a range of technological, procedural, and organizational components designed to ensure the confidentiality, integrity, and availability of government information and services. Here's a discussion on the key aspects of e-government security architecture:

- 1. Risk Assessment and Management:** The first step in designing an e-government security architecture involves identifying potential risks and vulnerabilities. This includes conducting regular risk assessments to evaluate the security



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includes conducting regular risk assessments to evaluate the security posture of government systems and prioritizing security measures based on the level of risk.

2. **Policy and Compliance:** Governments need to establish security policies and standards that govern the use and protection of electronic systems and data. These policies should align with relevant regulations and standards and ensure compliance with legal requirements regarding data privacy and security.
3. **Access Control:** Access control mechanisms are critical for limiting access to government systems and data only to authorized users. This includes implementing strong authentication methods, such as multi-factor authentication, and role-based access control to enforce the principle of least privilege.
4. **Data Encryption:** Encryption is essential for protecting sensitive data transmitted over networks and stored in government databases. Governments should implement strong encryption algorithms to secure



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databases. Governments should implement strong encryption algorithms to secure data both at rest and in transit, reducing the risk of unauthorized access or interception.

5. **Network Security:** Robust network security measures are necessary to defend against cyber threats such as malware, phishing attacks, and denial-of-service (DoS) attacks. This includes deploying firewalls, intrusion detection/prevention systems, and network segmentation to isolate sensitive systems and data from potential threats.
6. **Secure Software Development:** Governments should adopt secure software development practices to minimize the risk of vulnerabilities in e-government applications and systems. This includes conducting regular code reviews, implementing secure coding guidelines, and performing thorough security testing before deploying software into production.
7. **Incident Response and Recovery:** Despite preventive measures, security incidents may still occur. Governments need to have robust incident response plans in place to detect, contain, and mitigate the impact of



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incidents. Conducting forensic analysis, and



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may still occur. Governments need to have robust incident response plans in place to detect, contain, and mitigate the impact of security breaches promptly. This includes establishing procedures for reporting incidents, conducting forensic analysis, and restoring affected systems and data.

8. **Continuous Monitoring and Auditing:**

Continuous monitoring of government systems and networks is essential for detecting and responding to security threats in real-time. This involves implementing security information and event management (SIEM) systems, conducting regular security audits, and maintaining logs for compliance and forensic purposes.

9. **Vendor Risk Management:**

Many governments rely on third-party vendors and service providers for e-government solutions. It's crucial to assess the security posture of vendors and establish contractual agreements that define security requirements and responsibilities to ensure the protection of government data and systems.

10. **User Awareness and Training:**



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Employees and citizens about security best



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9. **Vendor Risk Management:** Many governments rely on third-party vendors and service providers for e-government solutions. It's crucial to assess the security posture of vendors and establish contractual agreements that define security requirements and responsibilities to ensure the protection of government data and systems.
10. **User Awareness and Training:** Lastly, user awareness and training programs are essential for educating government employees and citizens about security best practices, such as password hygiene, safe browsing habits, and recognizing phishing attempts. Well-informed users are better equipped to protect themselves and contribute to overall cybersecurity efforts.



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Best Answer of the Question

Explain in brief about e-governance initiatives in Srilanka.

Expert approved answer

Sri Lanka has been progressively embracing e-governance initiatives to enhance government efficiency, transparency, and citizen services. Some of the notable e-governance initiatives in Sri Lanka include:

1. **e-Development:** Sri Lanka has launched various e-development initiatives aimed at leveraging information and communication technology (ICT) to drive socio-economic development. These initiatives focus on expanding internet connectivity, promoting digital literacy, and fostering innovation in the ICT sector.
2. **Government Information Center (GIC):** The Government Information Center serves as a centralized platform for citizens to access government services and information online. It provides a wide range of services,



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programs.



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Best Answer of the Question

government services and information online. It provides a wide range of services, including online forms, applications, and updates on government policies and programs.

3. **e-Citizen Services Portal:** The e-Citizen Services Portal enables citizens to access a variety of government services online, ranging from obtaining birth certificates and passports to paying taxes and utility bills. This portal enhances convenience and accessibility for citizens by reducing the need for physical visits to government offices.
4. **e-Procurement System:** Sri Lanka has implemented an electronic procurement system to streamline the government procurement process and improve transparency and accountability. The system allows vendors to submit bids electronically and facilitates efficient procurement management across government agencies.
5. **e-Justice System:** The e-Justice System aims to modernize the legal system in Sri Lanka by digitizing court processes and case management. This initiative enhances



[VIEW ALL ANSWERS](#)



proceedings through the use of technology.



QN Bank Solution



Best Answer of the Question

aims to modernize the legal system in Sri Lanka by digitizing court processes and case management. This initiative enhances access to justice, reduces case backlogs, and improves the efficiency of legal proceedings through the use of technology.

6. **Digital Identity System:** Sri Lanka has introduced digital identity initiatives to provide citizens with secure and reliable digital identities for accessing government services online. These digital identity systems facilitate authentication and verification processes, ensuring the security and integrity of online transactions.
7. **ICT Infrastructure Development:** The Sri Lankan government has made significant investments in ICT infrastructure development to expand internet connectivity and enhance digital infrastructure across the country. This includes initiatives to improve broadband connectivity, establish digital hubs, and promote ICT literacy in rural areas.
8. **Open Data Initiatives:** Sri Lanka has launched open data initiatives to promote transparency and accountability in



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Best Answer of the Question

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8. **Open Data Initiatives:** Sri Lanka has launched open data initiatives to promote transparency and accountability in government operations. By making government data publicly available in machine-readable formats, these initiatives facilitate data-driven decision-making, innovation, and economic growth.



[VIEW ALL ANSWERS](#)





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Best Answer of the Question

Write short notes on: (any two)

1. Data system infrastructure
2. Model of Digital governance
3. Security standards.

Expert approved answer

Data System Infrastructure:

- Data system infrastructure refers to the foundational components and architecture that support the storage, processing, and management of data within an organization or across systems.
- It includes hardware such as servers, storage devices, and networking equipment, as well as software components like databases, data warehouses, and data management systems.
- The infrastructure must be designed to handle the volume, velocity, and variety of



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Best Answer of the Question

- The infrastructure must be designed to handle the volume, velocity, and variety of data generated by modern digital systems, including structured and unstructured data.
- Key considerations in data system infrastructure design include scalability, reliability, performance, and security to ensure the integrity and availability of data for decision-making and operational processes.
- Organizations often employ a combination of on-premises infrastructure, cloud services, and hybrid solutions to meet their data processing and storage requirements efficiently.

Model of Digital Governance:

- The model of digital governance refers to the framework and approach adopted by governments to leverage digital technologies for governance and public service delivery.
- Digital governance encompasses policies, strategies, and practices aimed at utilizing ICT to improve government efficiency,



[VIEW ALL ANSWERS](#)



- Models of digital governance can vary



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Best Answer of the Question

strategies, and practices aimed at utilizing ICT to improve government efficiency, transparency, and citizen engagement.

- Models of digital governance can vary based on factors such as the level of ICT adoption, regulatory environment, and governance structures within a country.
- Common components of digital governance models include e-government initiatives, open data initiatives, digital identity systems, and cybersecurity measures.
- Effective digital governance models prioritize citizen-centric service delivery, data-driven decision-making, collaboration across government agencies, and public participation in policymaking processes.
- Digital governance models evolve in response to technological advancements, changes in citizen expectations, and emerging challenges such as cybersecurity threats and digital inequality.

Security Standards:

- Security standards are guidelines and best



[VIEW ALL ANSWERS](#)





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Best Answer of the Question

- Security standards are guidelines and best practices established to ensure the confidentiality, integrity, and availability of information and systems, particularly in the context of cybersecurity.
- Security standards provide a framework for organizations to implement robust security controls and processes to protect against various cyber threats and vulnerabilities.
- Common security standards include ISO/IEC 27001 for information security management systems (ISMS), NIST Cybersecurity Framework developed by the National Institute of Standards and Technology, and PCI DSS for securing payment card data.
- Industry-specific security standards may also exist, such as HIPAA for healthcare data or GDPR for personal data protection in the European Union.
- Compliance with security standards helps organizations demonstrate due diligence in managing cybersecurity risks and may be required for regulatory compliance or contractual obligations.



[VIEW ALL ANSWERS](#)



such as access control, encryption, incident



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Best Answer of the Question

- Common security standards include ISO/IEC 27001 for information security management systems (ISMS), NIST Cybersecurity Framework developed by the National Institute of Standards and Technology, and PCI DSS for securing payment card data.
- Industry-specific security standards may also exist, such as HIPAA for healthcare data or GDPR for personal data protection in the European Union.
- Compliance with security standards helps organizations demonstrate due diligence in managing cybersecurity risks and may be required for regulatory compliance or contractual obligations.
- Security standards typically cover areas such as access control, encryption, incident response, risk management, and security awareness training, providing a comprehensive approach to cybersecurity governance and risk management.



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